



Trinity Grammar School

Mathematics Department

2016

TRIAL EXAMINATION
HSC ASSESSMENT TASK 4

Year 12 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- BOSTES approved calculators for this course may be used
- A Reference Sheet for this course is supplied
- In Questions **11 – 16**, show all relevant mathematical reasoning and/or calculations
- Write your BOSTES Student Number (Year 12 HSC) or Name (Year 10 or 11) **and** your Class teacher on the question paper **and** on any answer sheets or writing booklets used to write your responses to the questions submitted
- If you do not attempt a question you **must** submit an answer sheet **or** writing booklet for that question clearly indicating **N/A** and your Student Number or Name.
- **HSC Assessment Weighting: 40%**

BOSTES Student Number

(Year 12 only)

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Class Teacher:

.....

Name:

(Year 10 or 11 only)

.....

Do NOT write solutions on this question paper. Any working on this question paper will NOT be marked.

Date of examination:

Thursday, 25 August 2016

Total marks – 100

Section I

10 marks

- Attempt all Questions
- Allow about **15** minutes for this section

Section II

90 marks

- Attempt all Questions
- Allow about **2 hours 45** minutes for this section

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Section I 10 marks

- Attempt Questions 1 – 10
 - Allow about 15 minutes for this section
 - Shade the correct response on the multiple-choice answer sheet for Questions 1 – 10
 - Each question is worth 1 mark
-

1 Let $p = \frac{2 \cos x}{\sqrt{y}}$ where $x = 45^\circ$, $y = 8192$. The value of p in scientific form correct to three significant figures is:

- (A) 1.560×10^{-2}
- (B) 1.56×10^{-2}
- (C) 1.562×10^{-2}
- (D) 15.6×10^{-2}

2 The graph of a function $f(x) = a(x - h)^2 + k$ intersects the y -axis at $(0, 5)$ and has its vertex at the point $(4, 13)$. The value of a , of h and of k are respectively:

- (A) $a = -\frac{1}{2}$, $h = -4$, $k = -13$
- (B) $a = -\frac{1}{2}$, $h = -4$, $k = 13$
- (C) $a = -\frac{1}{2}$, $h = 4$, $k = -13$
- (D) $a = -\frac{1}{2}$, $h = 4$, $k = 13$

3 If $g(x) = \log_e(3x - 4)$, the slope of the tangent to $y = g(x)$ at the point where $x = 2$ is:

- (A) $-\frac{3}{2}$
- (B) $-\frac{2}{3}$
- (C) $\frac{2}{3}$
- (D) $\frac{3}{2}$

- 4 Rudolf has been given \$500 by his mother for his 18th birthday. Rudolf plans to immediately deposit this amount in a bank which offers an interest rate of 6.00% p.a., compounded quarterly, for three years.

Calculate the interest earned from the bank.

- (A) \$95.51
- (B) \$97.81
- (C) \$131.24
- (D) \$134.49

- 5 Evaluate the arithmetic series: $S_{46} = -7 + (-9) + (-11) + \dots + (-97)$.

- (A) -2800
- (B) -2392
- (C) -2295
- (D) -2184

- 6 Evaluate: $\int_0^{\frac{\pi}{8}} \sin 8x \, dx$

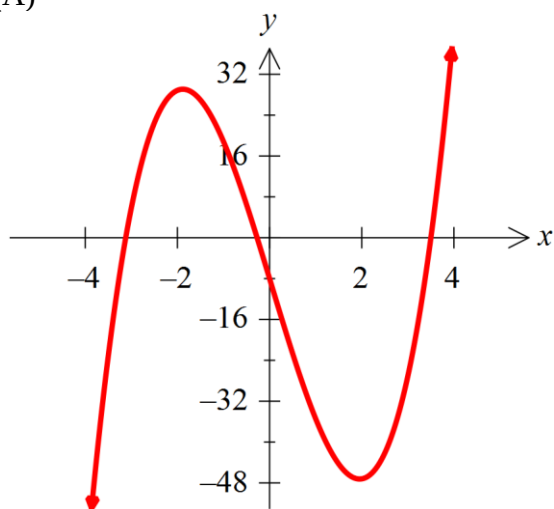
- (A) -2
- (B) $-\frac{1}{4}$
- (C) 0
- (D) $\frac{1}{4}$

- 7 A function $y = f(x)$ is differentiable on the interval $-\infty < x < \infty$ and has the following properties.

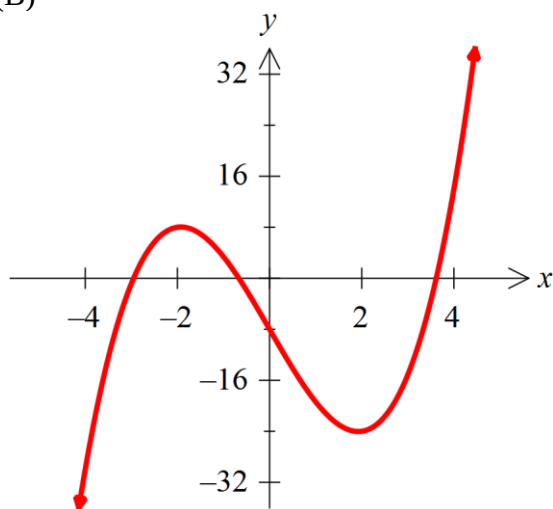
x	$y = f(x)$	Derivatives
$x < -2$		$y' > 0, y'' < 0$
-2	8	$y' = 0, y'' < 0$
$-2 < x < 0$		$y' < 0, y'' < 0$
0	-8	$y' < 0, y'' = 0$
$0 < x < 2$		$y' < 0, y'' > 0$
2	-24	$y' = 0, y'' > 0$
$x > 2$		$y' > 0, y'' > 0$

A possible graph of $y = f(x)$ could be:

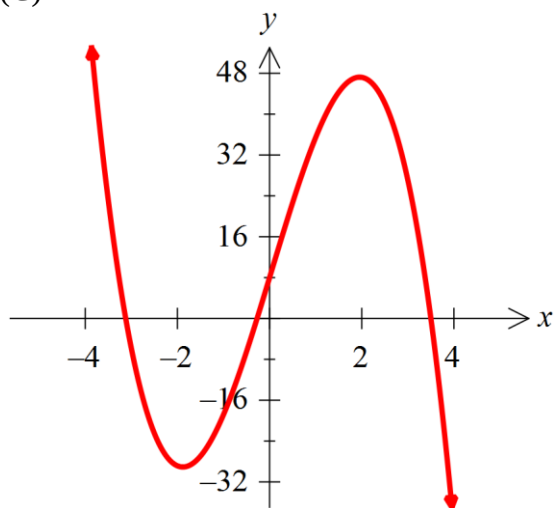
(A)



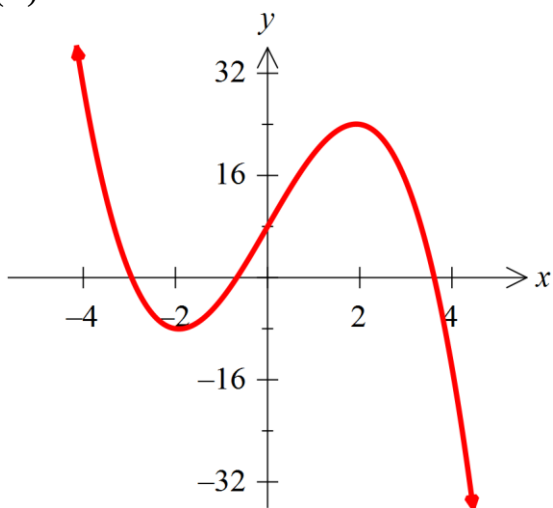
(B)



(C)



(D)



8 Solve for x if $\log_2 x + \log_2(x-2) = 3$.

- (A) $x = -4$ only
- (B) $x = -2$ only
- (C) $x = 4$ only
- (D) $x = -2$ and $x = 4$

9 When a camera flash goes off, the batteries immediately begin to recharge the flash's capacitor, which stores electric charge Q , given by $Q(t) = Q_0 \left(1 - e^{-\frac{t}{4}}\right)$. The maximum charge capacity is Q_0 and t is time measured in seconds.

How long does it take to recharge the capacitor to 90% of capacity?

- (A) $-4 \log_e \left(\frac{1}{10}\right)$ seconds
- (B) $-4 \log_e \left(\frac{9}{10}\right)$ seconds
- (C) $-4 \log_e \left(\frac{4}{9}\right)$ seconds
- (D) $-4 \log_e 10$ seconds

10 For what real values of c does the curve $f(x) = 5x^3 + cx^2 + 10x$ have a local maximum stationary point **and** a local minimum stationary point?

- (A) $c < -5\sqrt{6}$ or $c > 5\sqrt{6}$
- (B) $-5\sqrt{6} \leq c \leq 5\sqrt{6}$
- (C) $-5\sqrt{6} < c < 5\sqrt{6}$
- (D) $c \leq -5\sqrt{6}$ or $c \geq 5\sqrt{6}$

Section II 90 marks

- Attempt Questions 11 – 16
 - Allow about 2 hours and 45 minutes for this section
 - Answer each question in a SEPARATE writing booklet or answer sheet. Extra writing booklets are available.
 - In Questions 11 – 16, your responses should include all relevant mathematical reasoning and/or calculations.
-

Question 11 (15 marks)

Use a SEPARATE writing booklet

- (a) Solve for x if: $3 - (2 + 4x) = \frac{x}{2}$ 2
- (b) Write down the domain of the function $f(x) = \ln(x + 6)$. 1
- (c) Each day Stanley trains for a 10 km race. On the first day he runs 1000 m and then increases the distance by 250 m on each subsequent day.
- (i) On which day does Stanley run a distance of 10 km in training? 2
- (ii) Find the total distance Stanley will have run in training by the end of the day calculated in (c)(i), giving your answer in kilometres. 2
- (d) Solve the equation $\tan x = -\frac{1}{\sqrt{3}}$, for $0 \leq x \leq 2\pi$. 2
- (e) Calculate the size of each interior angle of a **regular** octagon? 1
- (f) A particle moves along a straight line so that its velocity, $v \text{ ms}^{-1}$, at time t seconds is given by $v = 240 + 20t - 10t^2$, for $0 \leq t \leq 6$.
- (i) Find the value of t when the speed of the particle is greatest. 2
- (ii) Find the acceleration of the particle when its speed is zero. 3

End of Question 11

Question 12 (15 marks)

Use a SEPARATE writing booklet

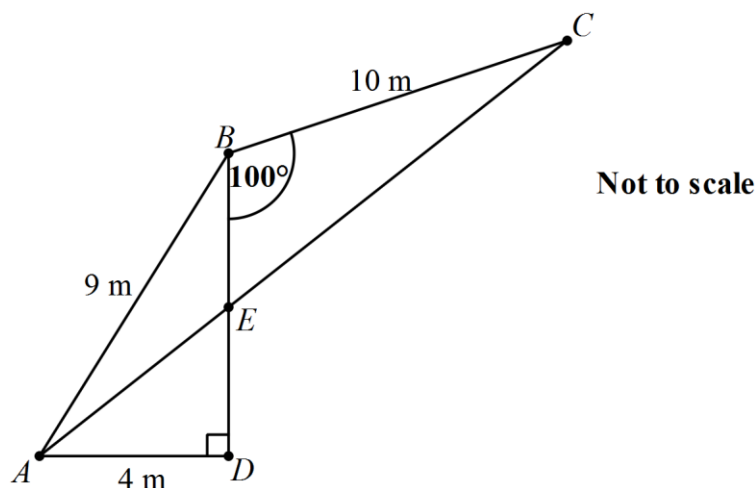
(a) Let $f(x) = |x-1|$ and $g(x) = |2x-1|$.

(i) Sketch, **on the same number plane**, the graphs of $y = f(x)$ and $y = g(x)$ 2

(ii) Hence or otherwise, solve $f(x) \geq g(x)$. 2

(b) Use Simpson's rule with **three** function values to estimate the value of $\int_0^4 \frac{dx}{1+\sqrt{x}}$. 2
Give your answer correct to two decimal places.

(c) In the diagram below, $AD = 4$ m, $AB = 9$ m, $BC = 10$ m, $\angle BDA = 90^\circ$ and $\angle DBC = 100^\circ$. The point E is the intersection of intervals AC and BD .



(i) Using $\triangle ADB$, find the size of $\angle ABC$, correct to one decimal place. 2

(ii) Hence, using the Cosine rule, find the length of AC , correct to the nearest integer. 2

(iii) Using the Sine rule, find the size of $\angle BAE$, correct to the nearest degree. 2

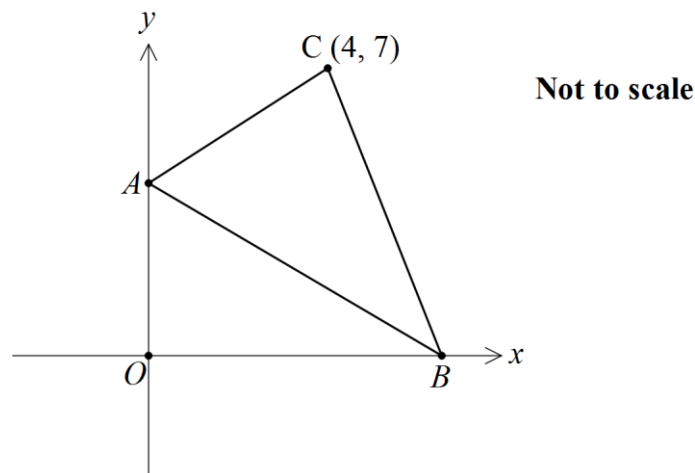
(d) Use the fact that $1 + \tan^2 x = \sec^2 x$, evaluate $\int_0^{\frac{\pi}{3}} \tan^2 x \, dx$. 3

End of Question 12

Question 13 (15 marks)

Use a SEPARATE writing booklet

- (a) Evaluate the infinite geometric series: $\sum_{n=1}^{\infty} \frac{5}{6^n}$ 2
- (b) A parabola is defined by the equation $y^2 - 4y + 4 = 24x$.
- (i) Write down the coordinates of the vertex. 1
- (ii) State the coordinates of the focus. 1
- (iii) Write down the equation of the directrix. 1
- (iv) Sketch the graph of the parabola showing the features found in part (b) (i), (ii) and (iii). 2
- (c) The diagram shows triangle ABC . Point C has coordinates $(4, 7)$ and the equation of line AB is $x + 2y = 8$.

**Copy this diagram into your writing booklet.**

- (i) Find the exact length of the interval AB (in the form $a\sqrt{b}$, where a and b are integers). 2

The point D lies on AB such that CD is perpendicular to AB .

- (ii) Find the equation of the line CD . 2
- (iii) Find the coordinates of the point D . 2
- (iv) Hence or otherwise, find the area of triangle ABC . 2

End of Question 13

Question 14 (15 marks)

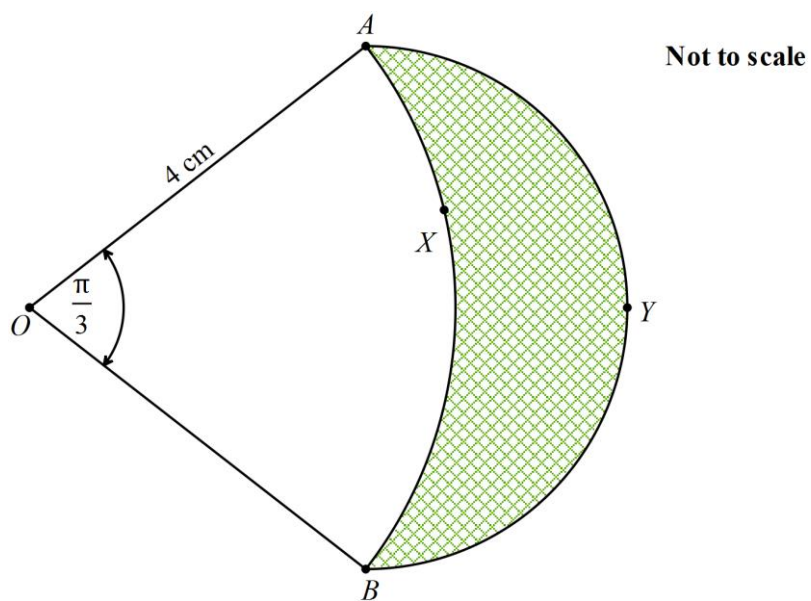
Use a SEPARATE writing booklet

- (a) For the function $f(x) = \frac{1}{4}x^3 - \frac{1}{2}x^2 - 8x + 1$,
- (i) find $f'(x)$ **and** $f''(x)$; 2
 - (ii) find the coordinates of the stationary points **and** determine their nature; 3
 - (iii) sketch the graph of $y = f(x)$, showing the features found in (ii); 2
 - (iv) for what value(s) of x is $f(x)$ concave up. 1
- (b) The population (P) of a bacteria culture is increasing at a rate determined by $\frac{dP}{dt} = kt$, $t \geq 0$ (hours). Initially $P = 11\,000$. After four hours has passed the population of the bacteria culture has tripled.
- (i) Verify that $P = P_0 e^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$. 1
 - (ii) Write down the value of P_0 . 1
 - (iii) Find the size of the bacteria culture after exactly 36 hours has passed. 3

Question 14 continues on page 11

Question 14 (continued)

(c)



In the diagram, AYB is a semicircle with AB as diameter and $OAXB$ is a sector of a circle with centre O and radius 4 cm . Angle AOB is $\frac{\pi}{3}$ radians.

Find the area of the shaded region.

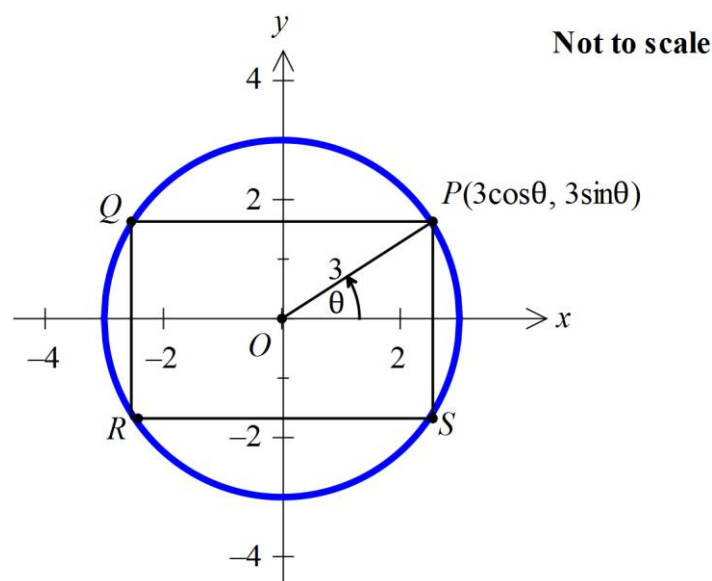
2

End of Question 14

Question 15 (15 marks)

Use a SEPARATE writing booklet

- (a) A rectangle $PQRS$ is inscribed in a circle of radius 3 cm and centre O , as shown below. All vertices of the rectangle lie on the circumference of the circle.



The point $P(3\cos\theta, 3\sin\theta)$ is a vertex of the rectangle and also lies on the circle.

The angle between interval OP and the x -axis is θ radians, where $0 \leq \theta \leq \frac{\pi}{2}$.

Let the area of rectangle $PQRS$ be A .

- (i) Show that $A = 36\sin\theta\cos\theta$. 2
- (ii) Use the product rule to show $\frac{dA}{d\theta} = 36(1 - 2\sin^2\theta)$. 2
- (iii) Hence, find the exact value of θ which maximises the area of the rectangle. 2
You do not need to test for maxima.

- (b) The acceleration $a \text{ ms}^{-2}$, of a particle at time t seconds is given by $a = 3\cos 2t$, for $t \geq 0$. The particle is at rest when $t = \frac{\pi}{4}$.

- (i) Write down the **next two** occasions (after $t = \frac{\pi}{4}$) when the particle is again at rest. 2
- (ii) Find the distance travelled (in metres), in the first $\frac{\pi}{2}$ seconds. Give your answer exactly in terms of π . 3

Question 15 continues on page 13

Question 15 (continued)

- (c) The depth, $h(t)$ metres, of water at the entrance to a harbour at t hours after midnight on a particular day is given by

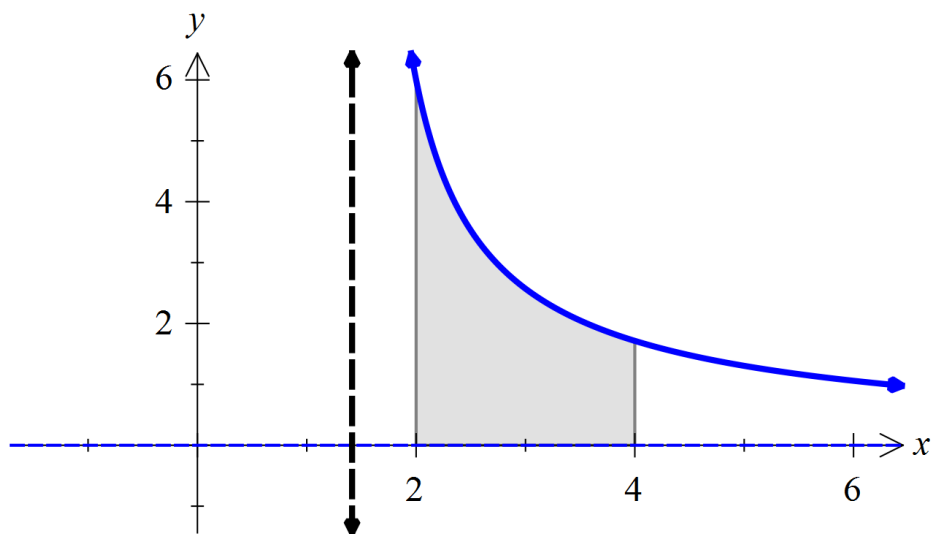
$$h(t) = 8 + 4\sin\left(\frac{\pi t}{6}\right), \text{ for } 0 \leq t \leq 24.$$

- | | | |
|-------|---|----------|
| (i) | Write down the amplitude of $h(t)$. | 1 |
| (ii) | Calculate the period of $h(t)$. | 1 |
| (iii) | Sketch the graph of $y = h(t)$ in the interval $0 \leq t \leq 15$. | 2 |

End of Question 15

Question 16 (15 marks)

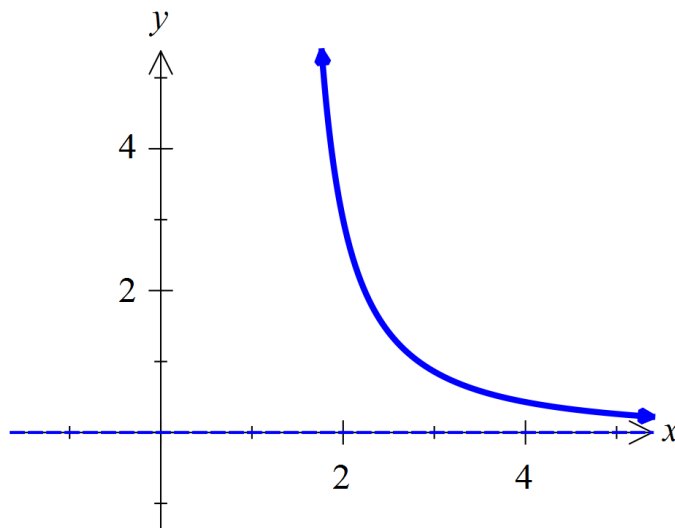
Use a SEPARATE writing booklet



- (a) The diagram above represents the graph of $y = \frac{6x}{x^2 - 2}$ for $x > \sqrt{2}$. 2

Find the area of the shaded region leaving your answer in the form $a \ln b$ where a and b are positive integers.

- (b)



- The diagram above represents the graph of $y = \frac{6}{x^2 - 2}$ for $x > \sqrt{2}$. 3

The region enclosed by the arc of the given curve from $y = 1$ to $y = 3$ and the y -axis, is rotated through 360° about the y -axis.

Find the volume of the solid formed. Leave your answer in terms of π .

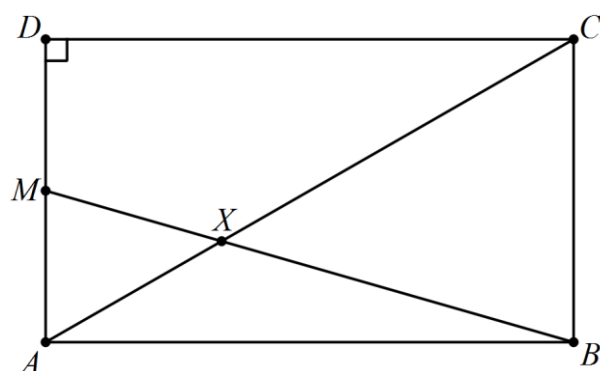
Question 16 continues on page 15

Question 16 (continued)

- (c) Janet has taken out a loan to pay for a well-earned holiday of \$250 000. The interest is calculated monthly at the rate of 6% p.a., compounded monthly. Janet agrees to repay the loan in full, at the end of each year, with interest, in ten equal annual instalments of \$ M . Let \$ A_n be the amount owing at the end of the n^{th} year just after the n^{th} repayment.

- (i) Show that an expression for A_2 in terms of M is given by 2
 $A_2 = 250\,000(1.005)^{24} - M(1 + 1.005^{12})$.
- (ii) Write down a similar expression for A_{10} . 1
- (iii) Calculate the value of M , correct to the nearest dollar. 1

(d)



Not to scale

In the diagram, $ABCD$ is a rectangle and $\frac{AD}{AB} = \frac{1}{2}$.

The point M is the midpoint of AD . The line BM meets diagonal AC at X .

Copy this diagram into your writing booklet.

- (i) Prove that $\triangle AXM \parallel \triangle BXC$. 2
- (ii) Prove that $\frac{CX}{AC} = \frac{2}{3}$. 2
- (iii) Prove that $\left(\frac{CX}{AB}\right)^2 = \frac{5}{9}$. 2

End of Question 16

End of paper

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Suggested Answer and Marking Scheme



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Mathematics Department

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BOSTES Student Number

(Year 12 only)

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Class Teacher:

EH

Name:

(Year 10 or 11 only)

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Date of examination:

Thursday, 25 August 2016

Total marks – 100

Section I

10 marks

- Attempt all Questions
- Allow about **15** minutes for this section

Section II

90 marks

- Attempt all Questions
- Allow about **2 hours 45** minutes for this section

1. *B*

2. *D*

3. *D*

4. *B*

5. *B*

6. *D*

7. *B*

8. *C*

9. *A*

10. *A*

Q11
HB

Q12
BL

Q13
MD

Q14
PK

Q15
JW

Q16
MM

Section I 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Shade the correct response on the multiple-choice answer sheet for Questions 1 – 10
- Each question is worth 1 mark

- 1 Let $p = \frac{2 \cos x}{\sqrt{y}}$ where $x = 45^\circ$, $y = 8192$. The value of p in scientific form correct to three significant figures is:

- (A) 1.560×10^{-2}
☒ (B) 1.56×10^{-2}
 (C) 1.562×10^{-2}
 (D) 15.6×10^{-2}

$$\frac{2 \cos 45^\circ}{\sqrt{8192}} = \frac{1}{64} = 0.015625 \text{ (exactly)}$$

- 2 The graph of a function $f(x) = a(x-h)^2 + k$ intersects the y-axis at $(0, 5)$ and has its vertex at the point $(4, 13)$. The value of a , of h and of k are respectively:

- (A) $a = -\frac{1}{2}$, $h = -4$, $k = -13$
 (B) $a = -\frac{1}{2}$, $h = -4$, $k = 13$
 (C) $a = -\frac{1}{2}$, $h = 4$, $k = -13$
☒ (D) $a = -\frac{1}{2}$, $h = 4$, $k = 13$

$$(h, k) = \text{Vertex} = (4, 13)$$

$$\begin{aligned} \text{sub } (0, 5) &\Rightarrow 5 = a(0-4)^2 + 13 \\ 5 &= 16a + 13 \\ 16a &= -8 \\ a &= -\frac{1}{2} \end{aligned}$$

- 3 If $g(x) = \log_e(3x-4)$, the slope of the tangent to $y = g(x)$ at the point where $x = 2$ is:

- (A) $-\frac{3}{2}$
 (B) $-\frac{2}{3}$
 (C) $\frac{2}{3}$
☒ (D) $\frac{3}{2}$

$$\begin{aligned} g'(x) &= \frac{3}{3x-4} \quad \left(D: x > \frac{4}{3} \right) \\ g'(2) &= \frac{3}{6-4} \\ &= \frac{3}{2} \end{aligned}$$

- 4 Rudolf has been given \$500 by his mother for his 18th birthday. Rudolf plans to immediately deposit this amount in a bank which offers an interest rate of 6.00% p.a., compounded quarterly, for three years.

Calculate the interest earned from the bank.

(A) \$95.51

(B) \$97.81

(C) \$131.24

(D) \$134.49

$$A = P(1 + i)^n$$

$$6\% \text{ p.a.} = \frac{6\%}{4} \text{ p.m.}$$

$$= \frac{0.06}{4}$$

$$= 0.015$$

$$n = 3 \times 4 = 12$$

$$\therefore I = 500(1.015)^{12} - 500 = \$97.81.$$

- 5 Evaluate the arithmetic series: $S_{46} = -7 + (-9) + (-11) + \dots + (-97)$.

(A) -2800

(B) -2392

(C) -2295

(D) -2184

$$S_n = \frac{n}{2}[a + l]$$

$$n = 46$$

$$a = -7$$

$$l = -97$$

$$\therefore S_{46} = \frac{46}{2}[-7 - 97]$$

$$= -2392.$$

- 6 Evaluate:

$$\int_0^{\frac{\pi}{8}} \sin 8x \, dx$$

(A) -2

(B) $-\frac{1}{4}$

(C) 0

(D) $\frac{1}{4}$

$$= -\frac{1}{8}[\cos 8x]_0^{\pi/8}$$

$$= -\frac{1}{8}[\cos \pi - \cos 0]$$

$$= -\frac{1}{8}[-1 - 1]$$

$$= -\frac{1}{8} \times -2$$

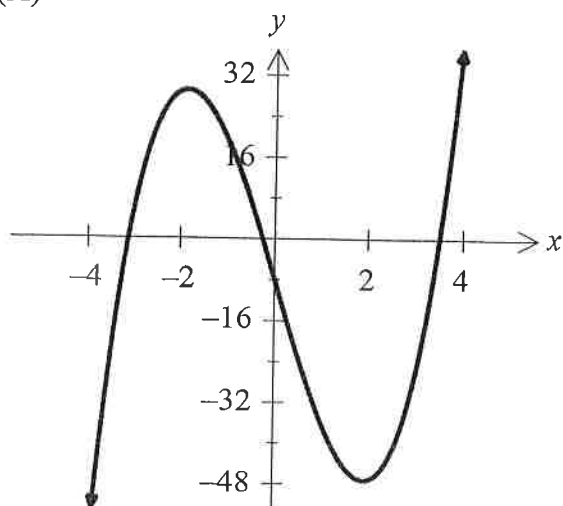
$$= \frac{1}{4}.$$

- 7 A function $y = f(x)$ is differentiable on the interval $-\infty < x < \infty$ and has the following properties.

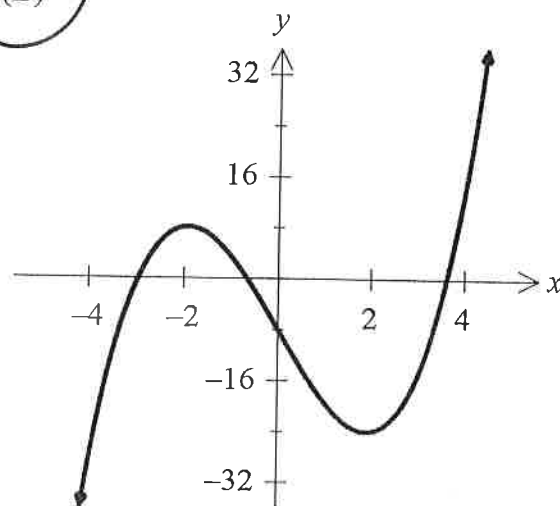
x	$y = f(x)$	Derivatives
$x < -2$		$y' > 0, y'' < 0$
-2	8	$y' = 0, y'' < 0$
$-2 < x < 0$		$y' < 0, y'' < 0$
0	-8	$y' < 0, y'' = 0$
$0 < x < 2$		$y' < 0, y'' > 0$
2	-24	$y' = 0, y'' > 0$
$x > 2$		$y' > 0, y'' > 0$

A possible graph of $y = f(x)$ could be:

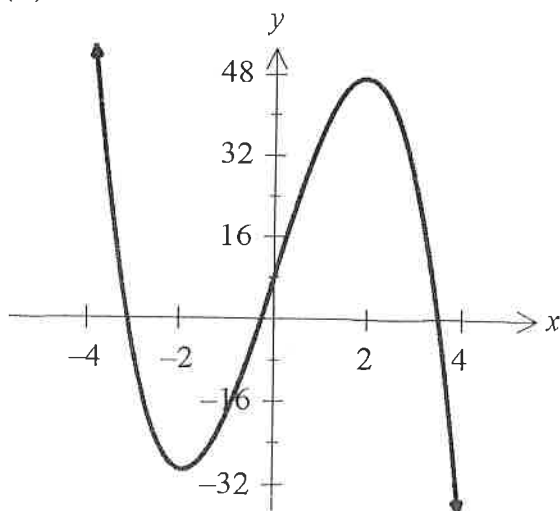
(A)



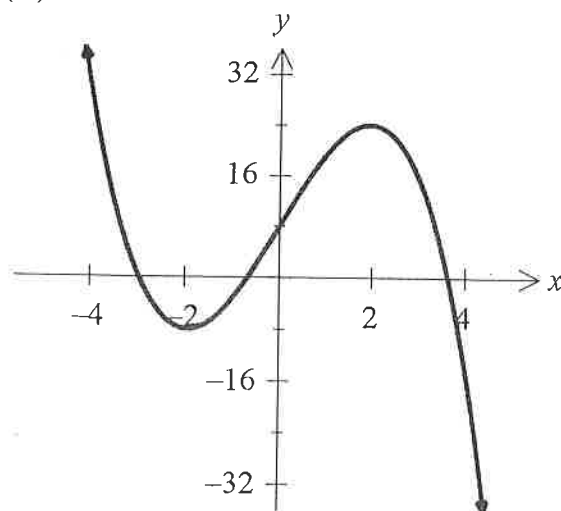
(B)



(C)



(D)



- 8 Solve for x if $\log_2 x + \log_2 (x-2) = 3$.

- (A) $x = -4$
 (B) $x = -2$
 (C) $x = 4$
 (D) $x = 2, x = 4$

$$\log_2 x(x-2) = 3$$

$$8 = x(x-2)$$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$x = 4, x = -2 \text{ (domain)}$$

- 9 When a camera flash goes off, the batteries immediately begin to recharge the flash's capacitor, which stores electric charge Q , given by $Q(t) = Q_0 \left(1 - e^{-\frac{t}{4}}\right)$. The maximum charge capacity is Q_0 and t is time measured in seconds.

How long does it take to recharge the capacitor to 90% of capacity?

(A) $-4 \log_e \left(\frac{1}{10}\right)$ seconds

(B) $-4 \log_e \left(\frac{9}{10}\right)$ seconds

(C) $-4 \log_e \left(\frac{4}{9}\right)$ seconds

(D) $-4 \log_e 10$ seconds

$$0.9 = 1 - e^{-t/4}$$

$$-\frac{t}{4} = \ln \frac{1}{10}$$

$$-t = 4 \ln \left(\frac{1}{10}\right)$$

$$\therefore t = -4 \ln \frac{1}{10}$$

- 10 For what values of c does the curve $f(x) = 5x^3 + cx^2 + 10x$ have a local maximum stationary point **and** a local minimum stationary point?

(A) $c < -5\sqrt{6}$ or $c > 5\sqrt{6}$

(B) $-5\sqrt{6} \leq c \leq 5\sqrt{6}$

(C) $-5\sqrt{6} < c < 5\sqrt{6}$

(D) $c \leq -5\sqrt{6}$ or $c \geq 5\sqrt{6}$

must have two roots to $f'(x) = 0$
 (not one only) $\therefore \Delta > 0$

$$f'(x) = 15x^2 + 2cx + 10$$

$$\Delta = 4c^2 - 4(15)(10)$$

$$4c^2 - 600 > 0$$

$$c^2 - 150 > 0$$

$$(c - 5\sqrt{6})(c + 5\sqrt{6}) > 0$$

Question 11:

a) $3 - (2 + 4x) = \frac{x}{2}$
 $1 - 4x = \frac{x}{2}$
 $2 - 8x = x$
 $9x = 2$
 $\therefore x = 2/9$

} ✓

✓

b) $f(x) = \log_e(x+6)$

D: $x > -6$

(accepted $-6 < x$)

✓

c) Using $T_n = a + (n-1)d$ since the sequence is arithmetic
 i) $10000 = 1000 + (n-1) \times 250$
 $n-1 = 36$
 $\therefore n = 37$ (37th day)

✓

✓

ii) Want S_{37} .
 Using $S_n = \frac{n}{2} [2a + (n-1)d]$

$S_{37} = \frac{37}{2} [2 \times 1000 + 36 \times 250]$
 $= 203,500 \text{ m}$
 $= 203.5 \text{ km}$

✓

✓

d) $\tan x = -\frac{1}{\sqrt{3}}$ ✓

Basic angle = $\pi/6$.

$\therefore$ Solution is $x = \pi - \pi/6$ and $x = 2\pi - \pi/6$

$\therefore x = 5\pi/6$ and $x = 11\pi/6$

① basic angle or 2 correct angles in degrees.

① ✓ 2 correct answers.

e) $n = 8 \therefore \text{angle} = \frac{180(8-2)}{8}$
 $= 135^\circ$

✓

f) $v = 240 + 20t - 10t^2$, $0 \leq t \leq 6$

i) speed is greatest when $\frac{dv}{dt} = 0$

$$\frac{dv}{dt} = 20 - 20t$$

$$\Rightarrow 20 = 20t$$
$$t = 1 \text{ second}$$

ii) Sub $v = 0$

$$\Rightarrow 0 = 240 + 20t - 10t^2$$

$$\Rightarrow 24 + 2t - t^2 = 0$$

$$\text{ie } t^2 - 2t - 24 = 0$$

$$(t-6)(t+4) = 0$$

$$\therefore t = 6 \text{ or } t = -4$$

$$\text{but } 0 \leq t \leq 6$$

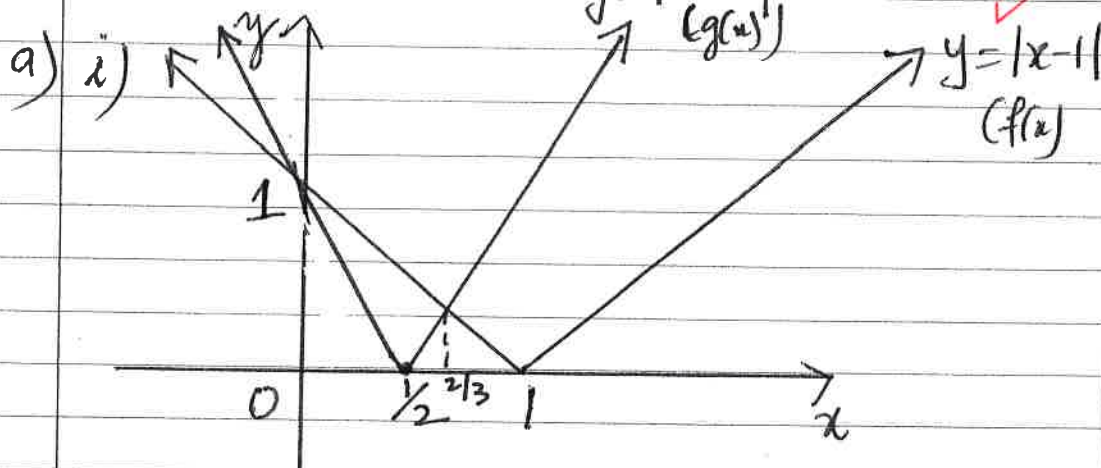
$$\therefore t = 6 \text{ only.}$$

$$\text{hence } a(6) = 20 - 20 \times 6$$

$$= -100 \text{ m/s}^2$$

Marker Comments:

Question 12:



ii) Hence: One solution is $x=0$ from above.
The other is obtained by solving

$$2x-1 = 1-x$$

ie' $3x = 2$

$$x = 2/3$$

$\therefore x=0$ or $x=2/3$ are critical values.

Hence $f(x) \geq g(x)$ when $\{x: 0 \leq x \leq 2/3\}$

Otherwise (one of several ways)

$$|x-1| \geq |2x-1|$$

$$(x-1)^2 \geq (2x-1)^2$$

$$(x-1)^2 - (2x-1)^2 \geq 0$$

$$\{x-1 - (2x-1)\}[x-1 + (2x-1)] \geq 0$$

$$(-x)(3x-2) \geq 0$$

$$x(3x-2) \leq 0$$

 $\Rightarrow \{x: 0 \leq x \leq 2/3\}$

b) $A \sim \frac{b-a}{6} \{y_1 + 4y_2 + y_3\}$ $y_2 = f\left(\frac{x_1+x_3}{2}\right)$

$$= \frac{4-0}{6} \left[\frac{1}{1} + 4 \times \frac{1}{1+\sqrt{2}} + \frac{1}{1+\sqrt{4}} \right]$$

$$= 1.99 \quad (\text{to 2 dp, no units required})$$

c) i) let $\angle ABD = \theta$.

$$\therefore \sin \theta = 4/9$$

$$\text{hence } \angle ABC = \sin^{-1}(4/9) + 100 \\ = 126.4^\circ \text{ (1 dp).}$$

ii) $AC^2 = 9^2 + 10^2 - 2 \times 9 \times 10 \cos 126.4^\circ$

$$\therefore AC \doteq 16.965 \dots \\ = 17 \text{ (nearest integer).}$$

iii) let $\alpha = \angle BAE$

$$\therefore \frac{\sin \alpha}{10} = \frac{\sin 126.4^\circ}{17}$$

$$\sin \alpha = \frac{10 \sin 126.4^\circ}{17}$$

$$\therefore \alpha = 28.2595 \dots \\ = 28^\circ \text{ (nearest degree)}$$

d) $\int_0^{\pi/3} \tan^2 x \, dx = \int_0^{\pi/3} \sec^2 x - 1 \, dx$

$$= \tan x - x \Big|_0^{\pi/3}$$

$$= \tan \pi/3 - \pi/3 - \tan 0 + 0$$

$$= \sqrt{3} - \pi/3. \quad (0.68 \rightarrow 2 \text{ dp})$$

approximate is penalised - 1 mark

Marking Comments.

ie. if an approximation is given in the final answer, a 1 mark penalty is applied.

"Evaluate $\Rightarrow$ exact answer"

Question 13:

$$a) = \sum_{n=1}^{\infty} 5 \times \left(\frac{1}{6}\right)^n$$

$$= \sum_{n=1}^{\infty} 5 \times \frac{1}{6} \left(\frac{1}{6}\right)^{n-1}$$

$$\therefore a = 5/6, r = 1/6$$

$$\begin{aligned} \text{hence } S_{\infty} &= \frac{a}{1-r} \\ &= \frac{5/6}{1-1/6} \\ &= 1. \end{aligned}$$

$$a = 5/6 \rightarrow \checkmark$$

✓

$$b) y^2 - 4y + 4 = 24x$$

$$(y-2)^2 = 24x$$

$$(y-2)^2 = 4(6)x \Rightarrow a = 6$$

$$i) V(0, 2)$$

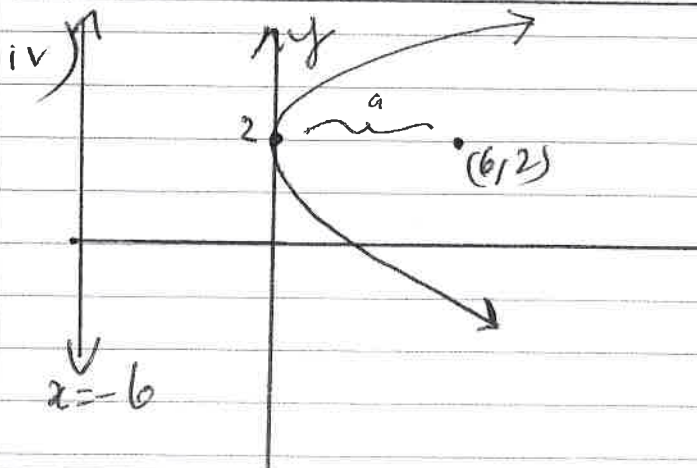
✓ must be a coordinate

$$ii) S(6, 2)$$

✓ must be a coordinate.

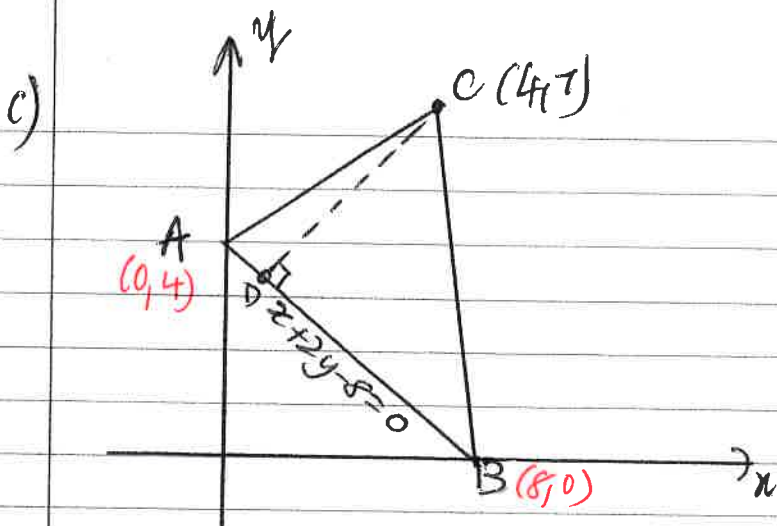
$$iii) x = -6$$

✓ must be an equation.



Correct shape
Correctly plotted } $\rightarrow$ 2
i), ii), iii)

② $\rightarrow$ from their i) ii) & iii)
if incorrect from above.



i) A and B are the y and x intercepts respectively.

$$\therefore \text{in } x + 2y - 8 = 0$$

$$\text{let } x = 0, \Rightarrow 2y = 8$$

$$y = 4$$

$$\text{let } y = 0 \Rightarrow x = 8$$

$$\therefore A = (0, 4) \text{ and } B = (8, 0)$$

$$\text{hence } AB = \sqrt{4^2 + 8^2}$$

$$= \sqrt{80}$$

$$= 4\sqrt{5} \text{ units}$$

ii) $m_{AB} = -1/2$

$$\therefore m_{CD} = 2$$

Using $y - y_1 = m(x - x_1)$ and $C(4, 7)$
Required equation of CD is:

$$y - 7 = 2(x - 4)$$

$$y = 2x - 8 + 7$$

$$y = 2x - 1 \text{ [or } 2x - y - 1 = 0]$$

iii) Solving $\begin{cases} 2x - y = 1 & \text{①} \\ x + 2y = 8 & \text{②} \end{cases}$

$$\text{in ②} \times 2 \Rightarrow 2x - y = 1 \text{ ①}$$

$$2x + 4y = 16 \text{ ③}$$

$$\text{③} - \text{①} \Rightarrow 5y = 15$$

$$\therefore y = 3$$

$$\text{hence } x = 2$$

$$\therefore D = (2, 3)$$

② $\rightarrow$ correct solution

① $\rightarrow$ use of avoid method obtaining at least one correct coordinate.

$$\begin{aligned}
 \text{iv) } A &= \frac{1}{2} \times CD \times AB \\
 &= \frac{1}{2} \times \sqrt{(-7-3)^2 + (4-2)^2} \times 4\sqrt{5} \\
 &= \frac{1}{2} \times \sqrt{20} \times 4\sqrt{5} \\
 &= 20 \text{ sq. units}
 \end{aligned}$$

Otherwise use $d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$ to calculate CD.

Marker Comments:

Question 14:

a) $f(x) = \frac{1}{4}x^3 - \frac{1}{2}x^2 - 8x + 1$

i) $f'(x) = \frac{3}{4}x^2 - x - 8$ ✓

$f''(x) = \frac{3}{2}x - 1$ ✓

① Provides correct derivative.

② Provides correct derivatives

ii) SP's occur when $f'(x) = 0$

$\Rightarrow \frac{3}{4}x^2 - x - 8 = 0$

$\Rightarrow 3x^2 - 4x - 32 = 0$

$(3x+8)(x-4) = 0$

$\Rightarrow \begin{cases} x = -8/3 & \text{or } x = 4 \end{cases}$

$\begin{cases} y = 91/27 & \text{or } y = -23 \end{cases}$

① Attempts to solve $\frac{dy}{dx} = 0$, or equivalent merit.

② Finds coordinates of both stationary points.

Nature: By the 2nd Derivative test:

$f''(-8/3) = \frac{3(-8/3) - 1}{2} = -5$

$-5 < 0$

$\therefore (-8/3, 91/27)$ is a local max. SP. ✓

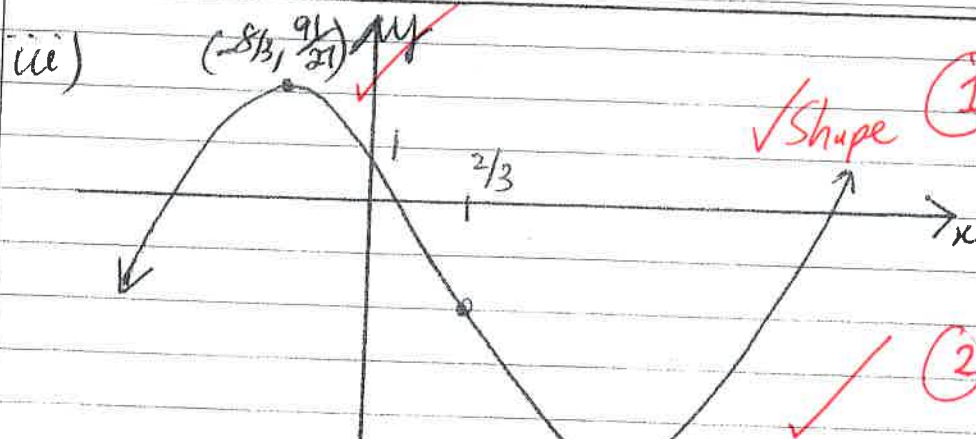
$f''(4) = \frac{3 \times 4 - 1}{2}$

$= 4$

> 0 ✓

$\therefore (4, -23)$ is a local min SP. ✓

③ Provides correct solutions



① Correctly plots and labels stationary points or equivalent merit. ✓ Shape

② Provides correct sketch

iv) $x > 2/3$ (except $x = 4/3$) $(4, -23)$ ✓

③ Provides correct solution

b) i) Given $P = P_0 e^{kt}$

$$\frac{dP}{dt} = k P_0 e^{kt}$$

$$= kP$$

① Provides correct solution

ii) $P_0 = 11,000$

① Provides correct value of P_0

iii) Want $P(36)$
 $P = 11000 e^{kt}$

Sub (4, 33000)
 $\Rightarrow 33 = e^{4k}$
 $4k = \ln 33$
 $\therefore k = \frac{1}{4} \ln 33$

① Finds the correct value of k

$\therefore P(36) = 11000 e^{36k}$
 $= 11000 e^{36 \left[\frac{1}{4} \ln 33 \right]}$
 $= 11000 \times 33^9$
 $= 216,513,000$

② Correct substitution.

③ Provides correct solution

c) Note $\triangle AOB$ is equilateral
 $\therefore AB = 4\text{cm}$ (diameter)

① Find the value of AB

Shaded area = Area of semi-circle - segment AOB
 $= \frac{1}{2} \times \pi \times (2)^2 - \frac{1}{2} \times 4^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$
 $= 2\pi - 8 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$
 $= \left(4\sqrt{3} - \frac{2\pi}{3} \right) \text{ sq. units}$

② Provides correct solution

Marker comments:

4.83 units m^2

Question 15:

a) i) $A = PQ \times PS$
 $= 2 \times 3 \cos \theta \times 2 \times 3 \sin \theta$
 $= 36 \sin \theta \cos \theta$

✓ clearly identifying a product 2 correct sides of the rectangle

ii) $\frac{dA}{d\theta} = 36 [\cos \theta \cos \theta + \sin \theta (-\sin \theta)]$
 $= 36 [\cos^2 \theta - \sin^2 \theta]$
 $= 36 [1 - \sin^2 \theta - \sin^2 \theta]$
 $= 36 [1 - 2\sin^2 \theta]$

✓ correct differentiation

✓ substitution of $(1 - \sin^2 \theta)$

iii) We want $\frac{dA}{d\theta} = 0$ for maxima/minima.

$\Rightarrow 1 - 2\sin^2 \theta = 0$

$\therefore \sin \theta = \pm \frac{1}{\sqrt{2}}$

but since $0 \leq \theta \leq \pi/2$ choose

$\sin \theta > 0 \therefore \sin \theta = \frac{1}{\sqrt{2}}$ only.

✓ valid method that leads to a correct answer.

Hence $\theta = \pi/4$ only.

(degree measure unacceptable).

✓ with justification that $\sin \theta > 0$ in $0 < x < \pi/2$.

b) $a = 3 \cos 2t$

i) Period of the motion is $\frac{2\pi}{2} = \pi$ seconds
 Rest at $t = \pi/4 \therefore$ the next two occasions would be:

$t = \pi/4 + \pi = 5\pi/4$ seconds.

and $t = \pi/4 + 2\pi = 9\pi/4$ seconds

✓ 1 valid method (e.g. by integration)

✓ [if bald answer given awarded 2 marks]

ii) $v = \frac{3}{2} \sin 2t + c$
 sub $(\pi/4, 0) \Rightarrow 0 = \frac{3}{2} \sin \pi/2 + c$
 $\therefore c = -3/2$

✓ correct answer from an incorrect method receives no marks.

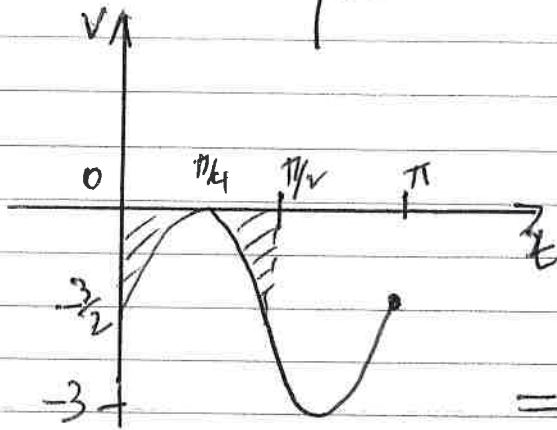
hence $v = \frac{3}{2} \sin 2t - \frac{3}{2} < 0$ for all t

$\therefore d = \left| \int_0^{\pi/2} \frac{3}{2} \sin 2t - \frac{3}{2} dt \right|$

✓ Recognise 1/2 is required and integral

ie $d = 2 \left| \int_0^{\pi/4} \frac{3}{2} \sin 2t - \frac{3}{2} dt \right|$

✓ → correct integration



$$= 2 \int_{\pi/4}^0 \frac{3}{2} \sin 2t - \frac{3}{2} dt$$

$$= 2 \left[-\frac{3}{4} \cos 2t - \frac{3t}{2} \right]_{\pi/4}^0$$

$$= 2 \left[\left(-\frac{3}{4} \cos 0 - 0 \right) - \left(-\frac{3}{4} \cos \frac{\pi}{2} - \frac{3\pi}{8} \right) \right]$$

$$= 2 \left[-\frac{3}{4} + \frac{3\pi}{8} \right]$$

$$= \left(\frac{3\pi}{4} - \frac{3}{2} \right) \text{ m}$$

✓ correct answer

[if $\frac{3}{2} - \frac{3\pi}{4}$ is given
a mark of 2 marks awarded]

c) $h(t) = 8 + 4 \sin\left(\frac{\pi}{6}t\right), 0 \leq t \leq 24$

i) $a = 4$ (metres)

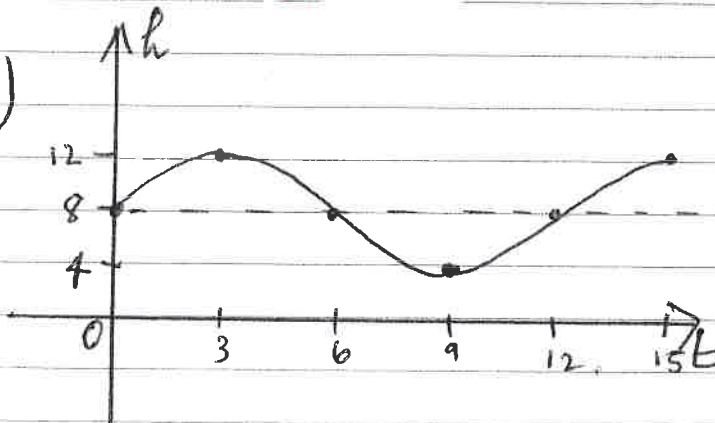
✓

ii) $T = \frac{2\pi}{\pi/6}$

$= 12$ hours.

✓

iii)



✓ Is shape

✓ points - position.

Marker Comments:

Question 1b:

a)
$$\text{Area} = \int_2^4 \frac{6x}{x^2-2} dx$$
$$= 3 \ln(x^2-2) \Big|_2^4$$
$$= 3 [\ln 14 - \ln 2]$$
$$= 3 \ln 7.$$

b) Making x^2 the subject:

$$yx^2 - 2y = b$$

$$\therefore x^2 = \frac{b+2y}{y} \quad (y \neq 0)$$

$$\text{ie } x^2 = \frac{b}{y} + 2.$$

$$\therefore V = \pi \int_1^3 \left(\frac{b}{y} + 2 \right) dy$$

$$= \pi [b \ln y + 2y]_1^3$$

$$= \pi [b \ln 3 + 6 - b \ln 1 - 2]$$

$$= \pi [b \ln 3 + 4] \text{ cubic units.}$$

c) i) $r = 6\% \text{ p.a.}$
$$= \frac{0.06}{12}$$
$$= 0.005$$

$$A_1 = 250\,000 (1.005)^{12} - M$$

$$A_2 = A_1 \times (1.005)^{12} - M$$
$$= [250\,000 (1.005)^{12} - M] (1.005)^{12} - M$$

$$A_2 = 250\,000 (1.005)^{24} - M (1 + 1.005^{12})$$

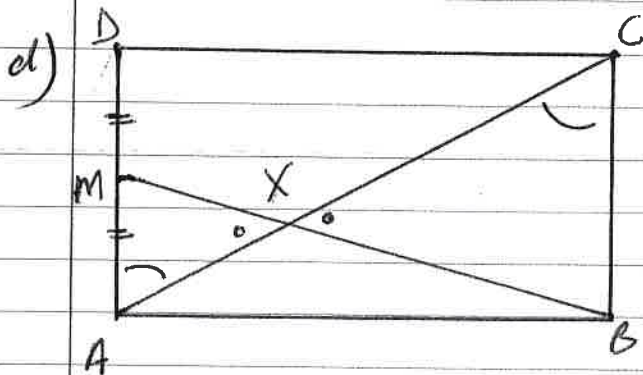
ii) $A_{10} = 250\,000 (1.005)^{120}$
$$- M (1 + 1.005^{12} + \dots + 1.005^{108})$$

iii) We want $A_{10} = 0$

$$\therefore M = \frac{250000 (1.005)^{120} [1.005^{12} - 1]}{[(1.005)^{120} - 1]}$$

$$= \$34,238$$

✓



i) In $\triangle AXM$ and $\triangle BXC$

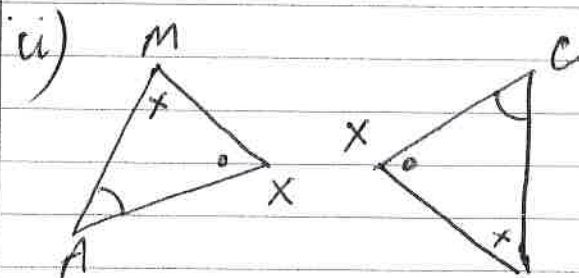
1. $\angle XMA = \angle XCB$ (vertically opp. angles)
2. $\angle XCB = \angle XMA$ (alt. angles in parallel lines $DC \parallel AB$).

$\therefore \triangle AXM \parallel \triangle BXC$ (equiangular)

✓ $\rightarrow$ 1 correct statement (= 1 mark)

$\rightarrow$ 3 correct lines (= 2 marks)

required.



In similar $\triangle$'s, corresponding sides are in proportion.

$$\therefore \frac{CX}{AX} = \frac{CB}{AM} = \frac{2}{1}$$

$$\therefore \frac{CX}{AX} = \frac{2}{1}$$

but $AX = AC - CX$

$$\therefore \frac{CX}{AC - CX} = \frac{2}{1} \Rightarrow CX = 2AC - 2CX$$

$\therefore 3CX = 2AC$

hence: $\frac{CX}{AC} = \frac{2}{3}$

✓

✓

iii) By Pythagoras' Theorem in $\triangle CAB$

$$AC^2 = AB^2 + CB^2$$

$$\text{but } CB = \frac{1}{2} AB \text{ (given)}$$

$$\therefore AC^2 = AB^2 + \frac{1}{4} AB^2$$

$$AC^2 = \frac{5}{4} AB^2$$

$$\text{since } AC = \frac{3CX}{2} \text{ from (ii)}$$

$$\text{Then } \left(\frac{3CX}{2}\right)^2 = \frac{5}{4} (AB)^2$$

$$\therefore \frac{9}{4} CX^2 = \frac{5}{4} AB^2$$

$$\left(\frac{CX}{AB}\right)^2 = \frac{5}{4} \times \frac{4}{9}$$
$$= 5/9$$

Marker Comments: